HighGen—Comparative genomics resources for cannabis

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Reference-guided whole-genome alignment pipelines are tuned for cross-species vertebrate panels. With **HighGen**, we adapted them in Galaxy across **217 *Cannabis sativa* assemblies** anchored on cs10—recalibrating repeat masking, empirical substitution scoring, chunked extension, and streaming 64-bit dynamic programming, paired with 43-genome Panagram *k*-mer bitmaps, a TOGA2 orthology pilot, and a 78-family phenotype ledger.

Galaxy

228-stanza (219-assembly) Hub
Attach depot.galaxyproject.org/highgen/hub.txt in the UCSC Genome Browser (Track Hubs → My Hubs).

A Four cross-species bottlenecks

In *Cannabis sativa*, interspersed repeats (~64–67% LTR retrotransposons) surround dense tandem metabolic arrays (Grassa et al. 2021, Lynch et al. 2025). Projecting 216 query assemblies onto the linear cs10 (CBDRx) reference coordinate grid confronts four vertebrate-default obstacles:

1 Masking bias
Arrived soft-masking spans 0.00–80.86% across 218 assemblies, and KegAlign (Gulhan et al. 2025) excludes lowercase seeds

2 Cross-species matrix
HOXD70 (Chiaromonte et al. 2002) targets below 70% identity, which in *Cannabis* is 49.7% chance noise (Ts/Tv = 0.872 vs 0.74 floor)

3 Superlinear scaling
Whole-chr LASTZ (Harris 2007) scales as $O(n^{1.9})$, while 217-way MULTIZ (Blanchette et al. 2004) overflows 32-bit scores and buffers MAFs in RAM

4 Macro-syntenic nets
Human-mouse netFilter -syn (Kent et al. 2003, maxFar = 200 kb) prunes rearranged tandem gene clusters

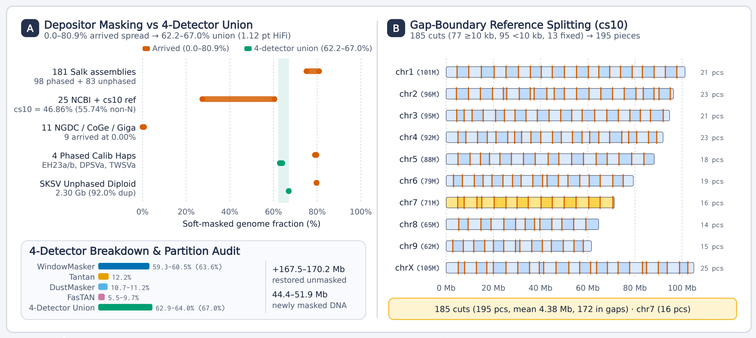


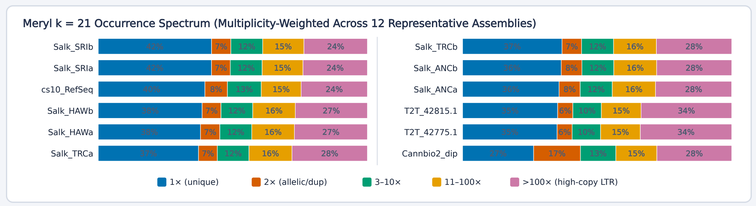
Figure 1 | Masking convergence and gap-boundary reference splitting. Stripping depositor masks across the 5-genome calibration panel and 5-genome intake pilot collapses a 0.0–80.9% intake spread into a 62.2–67.0% 4-detector union (**A**), while 185 chromosomal cuts (77 at N-runs ≥ 10 kb, 95 at N-runs under 10 kb, and 13 fixed 5 Mb cuts, wfi_cs10_splits.bed) partition cs10 into 195 ~5 Mb pieces plus 26 heavy unplaced scaffolds, mitochondrion NC_029855.1, and small.00, with mafSplit and maf_concat splitting and rejoining the 5 fixed-cut genes without base loss (**B**).

B Standardized 4-detector repeat masking

To eliminate submitter ascertainment bias across our 221 curated assembly records (217 non-redundant in MULTIZ— cs10 plus 216 queries representing 164 prefix groups and 156 distinct plants, scored with rosales_odb12 BUSCO, Lynch et al. 2025, Pike et al. 2025, 2026), our brc-tools UDT wrappers strip arrived masks and unite four de novo detectors (Morgulis et al. 2006, Frith 2011):

fasta-uppercase → WindowMasker DustMasker Tantan FasTAN → bedtools merge

Assembly / Cohort	Arrived Mask	4-Detector Union	Partition Audit and Compute
181 Salk assemblies (98 phased + 83 unphased)	75.12–80.86%	**62.19–67.04%** (5-genome intake pilot: 22.61 → 4.85 pt spread)	Depositors over-masked ~22% diverged unmasked DNA
25 NCBI GenBank + cs10	27.42–60.43% (cs10 46.86% total, 55.74% non-N)		Restricts cs10 seed nucleation from 53.14% to ~34–36%
11 NGDC / CoGe / GigaDB	0.00–0.68% (9 kept at 0.00%)		Prevents unmasked queries from seeding 100% of repeats
4 Phased Calibration Haps (EH23a/b, DPSVa, TWSVa)	**79.00–79.61%**	**62.90–64.02% (1.12-pt window)**	**WM 59.30–60.53%, Dust 10.7–11.2%, Tantan 12.2%, FasTAN 5.5–9.7% · 3.64–3.99 CPU-h/hap**
Diploid SKSV (2.30 Gb, 92.0% BUSCO dup)	79.71%	**66.95%** (4.05-pt panel window, WM 63.60%)	432.1 Mb restored, 139.1 Mb newly masked · 10.34 CPU-h



12-genome Meryl *k* = 21 occurrence spectrum: Phased haplotypes (cs10, 8 Pike et al. 2026 ONT, 2 IMPD T2T) carry 35–42% 1× and 6–8% 2× multiplicity-weighted *k*-mers, whereas uncollapsed diploid Cannbio-2 (Braich et al. 2020) carries 27% 1× and 17% 2×.

Across phased haplotypes, the union restores 167.5–170.2 Mb of depositor-masked DNA while newly masking 44.4–51.9 Mb—a net unmasked gain of +115.9 to 123.2 Mb per haplotype with exact lowercase FASTA ↔ 0-based BED parity.

C Empirical intra-species substitution matrix

We adapted the Chiaromonte et al. (2002) log-odds protocol across two independent phased haplotype pairs (EH23a × EH23b and DPSVa × TWSVa, 95.3% rosales_odb12 BUSCO completeness, 1.8–2.0% duplication). Inverting the training window to [85, 95)% identity, excising Salk GFF3 protein-coding CDS (–7.3% pre-chain segments), and enforcing single-chain colinearity (--max-chains 1) retained 54.5% of orthologous vs 6.6% median off-chromosome paralogous bases (**8.2-fold**) on Cb2 × cs10 and **18.0-fold** genome-wide on EH23a × EH23b.

82.76 Mb
Physical [85, 95)% pairs (331.05M 4×-symmetrized, Ts/Tv = 2.119, *H* = 1.4140 bits, Δ*H* = 0.0021)

9/9 Completed Arms
Wins vs HOXD70 (chr8 OOM, 9/10 loci $p = 0.0107$ vs $\alpha = 0.0083$) and 67/70 windows (95.7% [91.4, 100%])

5,033×
axtChain shuffled-null suppression (cs10 × H. lupulus) + scaled 682/51 gaps (79.6% CDS cov)

Normalized to max match 100 (+80 A/T, +100 C/G at 57.7% training AT, –92 transitions, –191 A-C/G-T, –197 C-G, –198 A-T transversions), our matrix also won **9/9 completed arms**, **10/10 loci**, and **62/70 windows** (88.6% [83.5, 94.6%]) vs unit ±100 and **55/70 windows** (78.6% [71.1, 86.2%], 9/10 loci) vs unchained across 40 false-match budgets.

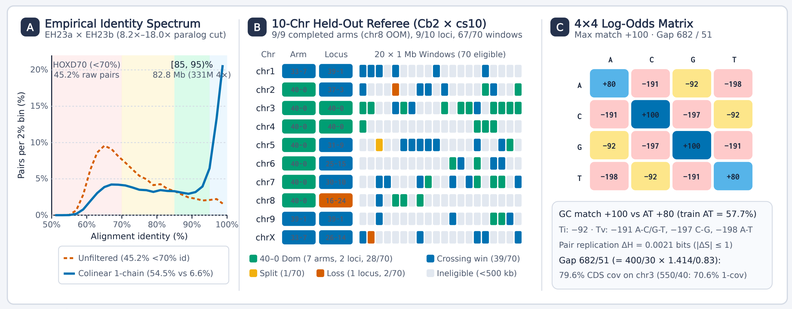


Figure 2 | Intra-species matrix calibration and 10-chromosome × 20-window held-out referee. Empirical identity spectra and single-chain colinearity separation (**A**), antisymmetric corner-threshold wins across 10 held-out Cb2 × cs10 chromosomes and 70 eligible 1 Mb windows (**B**), and symmetric 4×4 log-odds weights with 682/51 gap scaling (**C**).

D Collection-native LASTZ and streaming MULTIZ 11.4

- **Growler LASTZ (growler_ucsc_pair, LASTZ 1.04.52/1.04.60):** Couples KegAlign v0.3.4 GPU collection seeding (Gulhan et al. 2025) with indexed .2bit archives, compact LAV, and piped PSL chaining (Harris 2007, Kent et al. 2003). Batching anchor files into ≤50,000-segment chunks cuts EH23a × EH23b gapped extension **7.8-fold** (389.0 → 49.8 slot-h, 1.88 h wall-clock, median 28 min/job, under 3.97 GB RAM) while matching **99.99%** of UCSC's hg38 chr21 × panTro6 chr21 all.chain and rbest.chain pairs (56.8 vs 102.4 min) and producing byte-identical doRecipBest chains across 5 test queries.
- **Storage and 218-run production cohort:** Purging transient intermediate collections after workflow completion reduced total Galaxy history storage across all 218 runs (54,554 jobs, 10,325 job-hours) from **14.96 TB to 6.50 TB**—reclaiming **56.6%** (8.46 TB). Syntenic nets cover a median **65.2%** of cs10 non-N bases across 216 queries (range 12.2–85.1%, vs **70.0%** and 29.3–88.1% in rbest, with 40.0% diploid vs 54.0% phased median query-side coverage).
- **Streaming 64-bit column-count MULTIZ 11.3/11.4:** Replacing in-memory MAF buffering (v11.2) with coordinate-ordered streaming cursors and widening score accumulators to int64_t in v11.3 (multiz#19, fixing 29 deep blocks—24 with final scores ≥2^{31} —on NC_044376.1.02, Blanchette et al. 2004) cut hg38 5-way chrX memory **52.1-fold** (677 → 13 MiB, 217-way worker RAM ≤14.1 MB). Factoring mz_yama.c DP loops through column nucleotide counts (c_A, c_C, c_G, c_T, c_N) in v11.4 (multiz#28) accelerated a deep 217-way merge **32.8-fold** (953 → 29 s on NC_044372.1.24) and cut a 5 Mb piece runtime **6.5-fold** (26.8 → 4.1 core-h, with 82 chr pieces merged under v11.3 and 113 + scaffolds under v11.4) with byte-identical output.

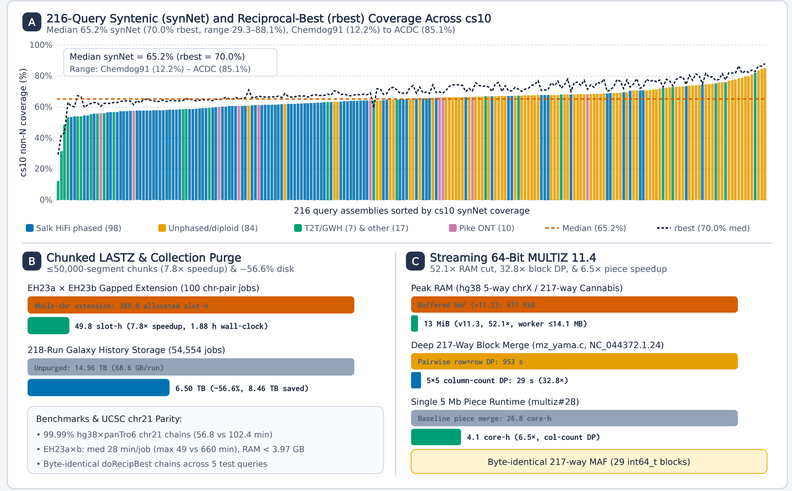


Figure 3 | 216-query syntenic coverage waterfall and scalability benchmarks. Sorted synNet (median 65.2%, range 12.2–85.1%) and rbest 1-to-1 (median 70.0%, range 29.3–88.1%) coverage across 216 queries by assembly class (**A**), chunked LASTZ speedup, hg38 chr21 parity, and 218-run storage purge (**B**), and MULTIZ v11.3/v11.4 streaming RAM (52.1×) and column-count DP speedup (32.8× block, 6.5× piece) (**C**).

E Tandem metabolic clusters and syntenic net limits

- **Genome-wide tandem census and permutation audit:** Adapting MCScanX tandem criteria (Wang et al. 2012) to stricter BLASTp thresholds ($E \le 10^{-10}$, top 10 hits), **3,035** of cs10's 25,296 RefSeq protein-coding genes reside in 1,208 tandem arrays ($K = 0$) and **5,123 (20.3%)** in 1,796 clusters ($K \le 10$), including **41 terpene synthases (TPS)** (31 in 9 clusters + 10 singletons) and **35 cannabinoid oxidocyclases (BBE-like)** (29 in 9 clusters + 6 singletons including CBDAS). Across 20,000 cluster-level permutations, nominal TPS low-identity enrichment (37.0% vs 11.0% controls, $p = 0.022$) does not survive family-wise correction ($p_{FWER} = 0.12$, threshold-free $p = 0.40$, BBE-like 2.8%, $p = 0.59$).
- **Functional CBDAS vs B-locus pseudogenes:** At cs10 functional CBDAS (LOC115697762, chr7:30.98 Mb, Grassa et al. 2021), our 217-way alignment separates **79 assemblies** (59 plants, including 56 Lynch full-length CBDAS) at ≥98.5% identity and **6 assemblies** below 93.0% (≥50% CDS) from **77 assemblies** (67 plants) at 93.0–98.5%—where all 63 Lynch-called contigs (62 THC-type, 1 both, 14 uncalled, Lynch et al. 2025) pair cs10 CBDAS with the ~94% B-locus pseudogene near THCAS.
- **netFilter -syn pruning vs paralog reuse:** Human-mouse distance and score cutoffs (maxFar = 200 kb, minTopScore = 300,000 accounting for 14.5% of losses) pruned **4.7% of TPS** (418/8,856: 4.2% clustered, 6.4% singleton) and **10.7% of BBE-like** (807/7,560: 7.5% clustered, 26.2% singleton vs 5.2%/13.1% controls)—dropping 54 CBDAS assemblies (under 50% CDS) in synNet, including Finola's intact 99.4% transposed copy (CM011610.1:21.84 Mb, 26.8 Mb from its enclosing gap). Simply removing netFilter -syn causes **46.4%** of recovered pairs (569/1,225 by interval, **455/1,225 = 37.1%** at ≥50% shared CDS bases) to reuse paralog query sequence, motivating complementary rbest nets and TOGA2 and TOAST/BOAST orthology (Hou et al. 2006, 2009).

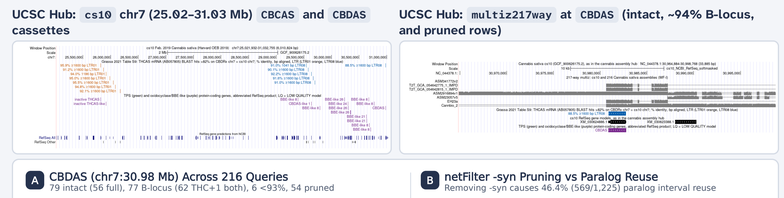


Figure 4 | Live HighGen UCSC Track Hub on chr7 and netFilter -syn trade-off. Browser views of the 6 Mb cannabinoid synthase super-cluster (left) and multiz217way at CBDAS (right), paired with 216-query CBDAS resolution (**A**) and netFilter -syn pruning vs 46.4% interval (37.1% CDS-base) paralog reuse (**B**).

F Panagram bitmaps, TOGA2 pilot, and phenotype ledger

- **Panagram v1.1.0 pan-*k*-mer indexing (panagram_index / panagram_view, Jenike et al. 2026, Kokot et al. 2017, Aylward et al. 2023):** Tier P1 (17 genomes: cs10 + 16 Salk haplotypes from 9 plants, 13.3 Gb) selected $k = 21$ over $k = 31$ (higher linear 1 Mb window Pearson r with rbest identity in **16/16 Salk haplotypes**, mean $r = 0.7150$ vs 0.6901, $\Delta r = +0.0249$ within the 0.05 equivalence band, 11.4× faster build). Tier P2 across **43 genomes** (32.64 Gb, 5.05 h on 16 Laila slots, 1.14 GB index) passed $r \ge 0.50$ in **41 of 42 chain-scored query genomes** ($r = 0.5848$–0.8118, including Pink pepper at $r = 0.7376$), passed non-PAR chrY separation (3/3 males 0.976–0.987 vs 3/3 V3 females ≤ 0.0773 across 205,619 100-bp-subsampled AH3Ma-absent positions, and 30/30 P4 XX chrY/chrX ≤ 0.320), flagged 2 higher-QV Guppy 5 SUP trio-binned ONT haplotypes on G3 count tolerance (Punto Rojo –10.4%, Cherry Pie –10.8%, Pike et al. 2025), and saw only one Sri Lankan ONT haplotype (ASM5191173v1, $r = 0.4763$, Merqury QV 24.01/24.11, Pike et al. 2026) fall below 0.50.
- **TOGA2 v2.0.9 orthology pilot (Kirilenko et al. 2023, Malovichko et al. 2026):** On the cs10 reference annotation, we stratified all **25,261** nuclear protein-coding genes (33,639 transcripts, excluding mitochondrion NC_029855.1) into **24,503 standard** and **758 reference-anomaly** genes alongside 221 inventoried all.chain datasets (1.45 TB: 116 VGP + 105 Laila across 218 query labels, 2 duplicate runs, and 1 calibration chain). On its zero-divergence identity-chain control (C-floor), TOGA2 called **25 of 24,503 lost/uncertain (0.10%**: 1 L + 24 UL against a ≤ 0.5% bar) in 54 min at 1.87 GB RAM, while completed Laila projections on EH23a called **2,077 (8.5%)** and **1,778 (7.3%)** standard genes L/UL on production vs loose chains (differing on 3.4% of genes, with the 40.1 h RepeatFiller+chainCleaner arm and B6 / B7 checks in progress).
- **Pedigree-verified phenotype ledger:** Joining 40 contributing sources—30 publications (29 paper texts, 2 supplements, and 988 Lynch et al. 2025 Figshare rows across 190 assemblies, from 43 audited paper extractions in passed.tsv) plus 10 NCBI BioSamples—across 215 assemblies yielded **1,341 trait records** across **154 plants in 78 pedigree families** (69 plants in the largest Salk family, 71 singletons). Under strict measured / reported filtering, 0 of 14 traits meet our ≥30-family, ≥2-class (≥5 families/class) association screen (Varin spans 30 families with HO40 in 3 classes and 37/38 non-WT plants in HO40, whereas Chemotype reaches 30 families with 2 classes ≥5 if Lynch's HPLC method sentence covers its column).

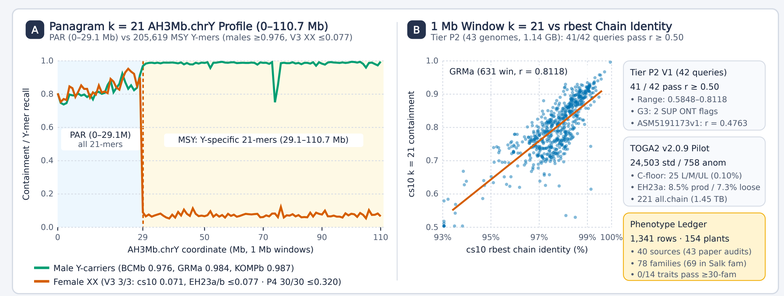


Figure 5 | Alignment-free Panagram chrY profile and rbest correlation. Chromosome-wide AH3Mb.chrY (0–110.7 Mb) *k* = 21 profile showing PAR all-21-mer containment (0–29.08 Mb) and MSY male-specific 21-mer recall across 205,619 subsampled positions (**A**), and all 631 eligible 1 Mb windows on GRMa ($r = 0.8118$) with Tier P2, TOGA2, and phenotype scorecards (**B**).

Open tools, workflows, and 228-stanza (219-assembly) browser hub

4-detector masking → brc-tools **GPU seeding + chunked LASTZ → growler_lastz_lav** **217-way streaming TBA → MULTIZ 11.4** **Pan-*k*-mer + ortholog → Panagram and TOGA2**

Availability: HighGen 228-stanza UCSC Track Hub (219 unique assemblies across 157 plants + 9 alias stanzas: multiz217way.bigMaf.bb, reading frames, pairwise chains/nets, rbest chains, gene models, and repeat masks): https://depot.galaxyproject.org/highgen/hub.txt (each assembly's hub page credits its source, with 9 CoGe/NGDC redistribution licences pending per assembly_licences.md). Analysis repository, scoring matrix, and scripts: https://github.com/richard-burhans/cannabis-genome. Galaxy tool wrappers (panagram_index, panagram_view, growler_lastz, gt-multiz-roast) and workflows (growler_ucsc_pair, wfi_split, wfi_roast, wfi_track): https://github.com/richard-burhans/galaxytools (MIT) and https://github.com/nekrut/brc-tools · MULTIZ 11.3/11.4: https://github.com/multiz/multiz · LASTZ 1.04.52/1.04.60: https://github.com/lastz/lastz · KegAlign 0.3.4: https://github.com/galaxyproject/KegAlign · Panagram 1.1.0: https://github.com/kjenike/panagram · TOGA2 2.0.9: https://github.com/hillerlab/TOGA2.

HighGen Hub · Galaxy Tools · Repository